

An Underactuated Dual-Slide Adaptive Robotic Hand with Strong Environmental Adaptability *

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Abstract—This paper proposes a grasping system based on a linear slide rail mechanism, characterized by strong environmental interaction and high adaptability. Its core concept involves coupling the linear slide rail with the finger plate of the robotic hand through a slot mechanism to achieve robust environmental adaptability and stable gripping during the grasping process. Through unique innovation, we have designed the DS Hand as its concrete implementation. Compared to other robotic hands with different complex mechanisms, this robotic hand features a simpler structure while possessing exceptional capabilities in passive adaptation, multi-mode grasping, and clamping. Theoretical analysis verifies its strong self-adaptability and experimental results demonstrate that the system can achieve stable grasping and holding of objects with various shapes and surface roughness. The proposed linear mechanism offers significant industrial applicability and low manufacturing cost. It does not rely on complex linkage-based linearization mechanisms and enables large-scale adaptive grasping. This work provides a low-cost and efficient solution for robotic grasping and clamping systems.

Keywords—Underactuated robotic hand, Adaptive grasp, Environmental adaptability, Adaptive switching between grasping and scooping

I. INTRODUCTION

Robotic grasping technology, as a critical link in robot-environment interaction, directly determines the operational capabilities of robots in complex scenarios such as industrial automation, logistics sorting, and space exploration. However, the high uncertainty in the shape, size, and surface properties of target objects in unstructured environments poses severe challenges to the adaptability, reliability, and cost-effectiveness of grippers [1].

To address these challenges, research has evolved primarily along two directions: one is the pursuit of high-degree-of-freedom dexterous hands, such as the Utah/MIT Hand [2], DLR Hand [3], and Stanford/JPL Hand [4], which, although capable of fine manipulation, suffer from complex control systems and high manufacturing costs; the other is the development of underactuated and adaptive grasping mechanisms, aiming to achieve passive adaptation through mechanical intelligence with fewer actuators than degrees of freedom, thereby striking a balance between performance and cost [5].

Underactuated and adaptive grippers have become the forefront of current research. Representative work includes tendon-driven designs like the Barrett Hand [6], the SARAH Hand [7]—the first underactuated hand used on the International Space Station—and innovative designs based on material properties, such as the universal gripper utilizing the jamming effect of granular materials [8] and the gripper employing magnetorheological fluid [9]. To further enhance grasping performance for specific targets like sheet-like objects, researchers have introduced mechanisms capable of generating linear trajectories, such as the Chebyshev linkage [10], Hoeckens mechanism [11], and Peaucellier linkage [12], to overcome the arc problem inherent in traditional rotational grasping trajectories and achieve stable parallel pinching.

Despite significant progress, existing designs still face limitations in handling ultra-thin objects, adapting to stepped environmental constraints, or achieving rapid and stable grasping mode switching, often due to mechanism complexity, insufficient adaptability, or high cost. For instance, the Omega Hand [13] experiences performance degradation under table constraints, while some linear grippers based on linkages [14] require improvement in achieving strong environmental adaptability and stable gripping. Furthermore, theoretical modeling and analytical frameworks for underactuated grasping have provided essential support for the development of this field [5].

Addressing the aforementioned problems and challenges, this paper proposes an underactuated adaptive robotic hand based on a dual-linear-slide mechanism. The core innovation of this design lies in utilizing a parallel dual-linear-slide mechanism to generate precise linear trajectories, combined with an underactuated transmission mechanism, enabling automatic switching between parallel pinching and environmentally adaptive grasping modes. The dual-slide design significantly enhances the stability of the grasping process and the adaptability to ultra-thin objects and stepped constraints. Secondly, through the optimization of the transmission chain and idle stroke, efficient mode switching with low actuation requirements is achieved. Finally, the overall structure strives for simplicity, aiming to control manufacturing costs while ensuring high performance, thereby improving feasibility for industrial deployment. This design is named the Dual-Slide Hand (DS Hand).

The structure of this paper is as follows: Section II introduces the mechanical design and working principle of the hand; Section III presents the kinematics analysis, static analysis and performance analysis; Section IV validates its grasping performance through experiments; Section V summarizes the full text and outlines future work.

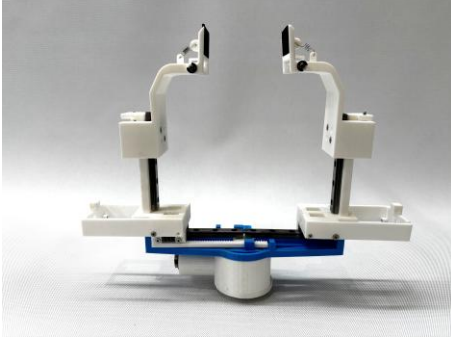


Fig. 1. The DS Hand

II. DESIGN OF THE DS HAND

This section elaborates on the mechanical structure of the DS Hand and how its design achieves the various intended functional characteristics with each feature explained in turn.

A. Design for Environmental Adaptability

The linear slideway mechanism is the core of this design. A dual linear slideway is installed on the horizontal base, with the motion trajectory of the slider arranged perpendicular to the grasping plane to achieve linear displacement. Upon collision with the grasping plane, the fingertip provides significant buffering, effectively adapting to the external environment, protecting the hand structure, and keeping the contact forces within a certain limit. This configuration enhances grasping stability and environmental adaptability, while also laying the foundation for the scooping function. Furthermore, through meticulous design, the horizontal offset of the motion trajectory relative to the fixed base is carefully adjusted to prevent internal interference and collisions caused by excessive or insufficient offset. This achieves structural simplicity and low manufacturing cost while eliminating internal interference and instability.

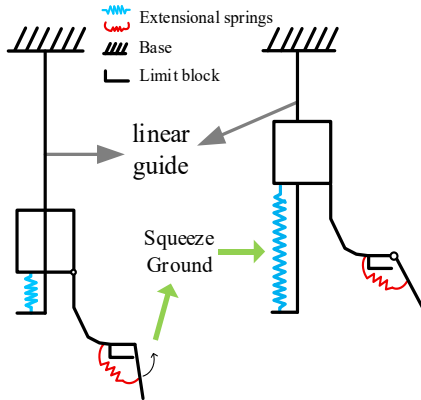


Fig. 2. Comparison of the structures before and after squeezing the ground

Fig. 2 displays the strong environmental interaction capability of the DS Hand after colliding with the ground. As shown in Fig. 5, A-B-C-D-E-F constitutes an integrated rigid structure, through which external forces are transmitted along the path $G \rightarrow F \rightarrow$ the rigid entirety, thereby avoiding the dispersion of force flow and local stress concentration. Rigid structures ensure that all geometric parameters remain constant during loading, thereby enhancing motion repeatability accuracy. The vertical displacement of the slider, together with the finger plate, exceeds 55 mm, offering a wider range of vertical adaptation compared to mechanisms that also have a linear trajectory. Fig. 3 illustrates the installation of the unilateral linear slideway, the spring mounting positions, and the slot design. The unique design of the spring locations and the slots ensures stable vertical displacement of the slider and the finger plate without deviation. The slot design allows the finger plate to move up and down smoothly while greatly simplifying the overall structure. This simplified layout not only improves space utilization but also increases the horizontal width of objects that can be grasped, enabling the DS Hand to handle larger objects. A detailed kinematic analysis and force analysis of the connecting slider linkage will be presented in subsequent sections.

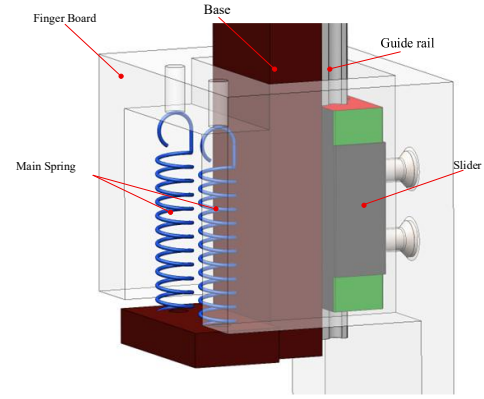


Fig. 3. The position of the slot and springs

B. Design of Adaptive Shoveling

The DS Hand achieves adaptive scooping through a stop plate installed on the finger plate base and a spring connected between the finger plate and the finger plate base (Fig. 4).

When the robotic hand contacts the grasping plane, the spring and the stop plate work in concert. The spring provides a clockwise torque, while the stop plate offers a unidirectional rotational constraint, ensuring the finger plate remains perpendicular to the grasping plane. As the applied pressure increases, the finger plate begins to rotate, gradually aligning itself parallel to the grasping plane. Throughout this process, the spring supplies a progressively increasing clockwise torque, maintaining the finger plate in a tense state. When the finger plate is subjected to excessive vertical pressure, it becomes completely parallel to the grasping plane. This fully demonstrates its adaptability and stability, as it remains secure and undamaged even under over-pressurization conditions. This structure also enables the scooping function. Pressing down on one side of the robotic arm changes the orientation of the finger plate, engaging the scooping state. Subsequent section will

demonstrate the process of the DS Hand successfully scooping up a card.

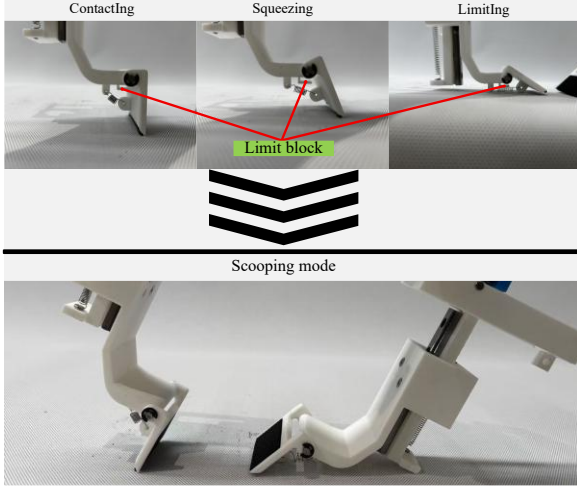


Fig. 4. The implementation principle of shoveling

C. Design of Automatic Switching between Adaptive Pinching and Scooping

When no vertical pressure is applied to the finger plate, the DS Hand remains in the horizontal pinching state. Once vertical pressure is applied to one side of the finger plate, it rotates under the guidance of the spring, transitioning into the scooping state (Fig. 4). When the pressure is released, the spring applies a torque, causing the finger plate to rotate back to its original position until it contacts the stop plate. At this point, the finger plate is again perpendicular to the grasping plane, returning to the pinching state. This design preserves the rotational degree of freedom, laying the mechanical foundation for the automatic switching between pinching and scooping functions.

Based on the aforementioned process, the DS Hand successfully achieves adaptive transition between horizontal pinching and scooping actions.

III. THEORETICAL ANALYSIS

A. Kinematic Analysis

In the following, the derivation is based on Fig. 5.

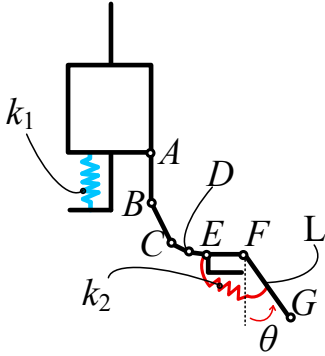


Fig. 5. The detailed structural diagram of the DS Hand

When the material thickness is not considered, the two actual springs connecting the slider can be abstracted as an equivalent

spring with a stiffness coefficient of k_1 . We focus on the relationship between the displacement of point G and the geometric parameters of the mechanism. Let point F be the center of rotation, the length of link FG be L , the length of L is 39 mm, and the angle of rotation be θ (taking the vertical direction as the reference). The displacement vector of point G can be expressed as

$$\vec{r}_G = \vec{r}_F + L(\sin \theta \hat{i} + \cos \theta \hat{j}) \quad (1)$$

In this context, $\vec{r}_F$ is the displacement vector of the rigid body $A-B-C-D-E-F$. Since $A-B-C-D-E-F$ forms a rigid body, its motion can be simplified as planar motion, with the constraint that the distances between all points remain constant

$$\|\vec{r}_A - \vec{r}_B\| = c, \|\vec{r}_B - \vec{r}_C\| = c \dots \quad (2)$$

The vertical displacement component of point G can be further simplified as

$$y_G = y_F + L \sin \theta \quad (3)$$

Where y_F is the vertical displacement of the rigid body, When the external force is applied to the entire hand:

$$y_F = \frac{F_{\text{ext}}}{k_1} \quad (4)$$

The kinematic equations indicate that the displacement of point G results from the superposition of the rigid body's translational motion and its local rotation. Furthermore, the rate of change of θ directly influences the velocity of point G .

Assuming the external force $F_{\text{ext}} = 10$ N, and based on the spring stiffness $k_1 = 0.5$ N/mm, without considering the limiting effect of spring k_2 on the force applied to the finger plate, the analysis of y_G is performed, resulting in Fig. 6.

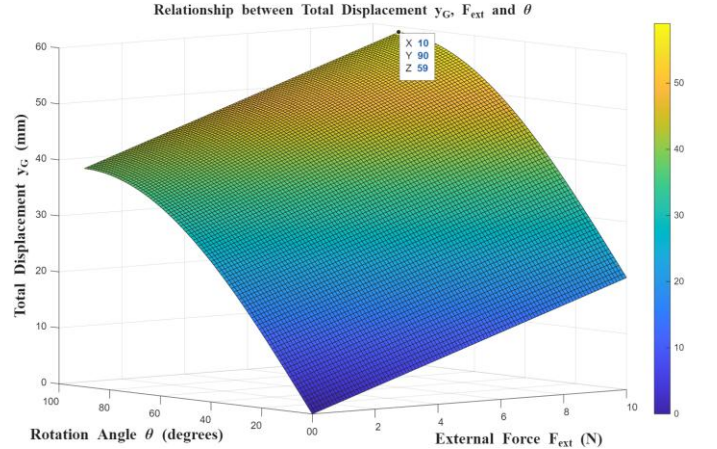


Fig. 6. The relationship between the external force, the rotation angle, and the displacement of point G

As shown in Fig. 6, under the action of a 10 N external force, when the angle θ reaches the 90-degree rotational limit, the displacement of point G can reach 59 mm, fully demonstrating its environmental adaptability. When subjected to external forces, the DS Hand exhibits good adaptability.

Derivation through kinematic formulas demonstrates that the DS Hand structure possesses a high degree of kinematic determinism. The integrated rigid-body design ensures that the mechanism undergoes no internal deformation during motion, with the relative positions of all points strictly maintained.

B. Static Analysis

The static analysis is based on the principle of virtual work, considering the upward external force F_{ext} generated at point G during table contact. The virtual work equation for the system can be written as

$$\delta W = F_{ext} \cdot \delta y_G - \sum_{i=1}^2 k_i \Delta L_i \cdot \delta(\Delta L_i) = 0 \quad (5)$$

Where k_i and ΔL_i represent the stiffness and deflection of the tension spring, respectively (where $i = 1, 2$). The equation shows a direct relationship between the external force F_{ext} and the elastic deformation of the springs ($\Delta L_1, \Delta L_2$). This means that when an external force is applied, the system transmits and adjusts these forces through the deformation of the springs, ensuring the stability and control capabilities of the robotic hand. Due to the constraints of the rigid body, the virtual displacement δy_G is coupled with $\delta \theta$, and their relationship is given by

$$\delta y_G = \delta y_F - L \sin \theta \delta \theta \quad (6)$$

Substituting the equilibrium conditions and utilizing the geometric relationship $\Delta L_i = L_i(\theta) - L_{i0}$ (where L_{i0} is the free length of the spring), the force equilibrium equation can be obtained as

$$F_{ext} = k_1 \frac{\partial(\Delta L_1)}{\partial y_G} \Delta L_1 + k_2 \frac{\partial(\Delta L_2)}{\partial y_G} \Delta L_2 \quad (7)$$

Static formulas reveal that the DS Hand possesses superior force control capabilities. The external force F_{ext} is uniformly distributed through the rigid body, avoiding stress concentration and enhancing structural reliability.

C. Performance Analysis

The overall performance synthesizes the kinematic and static results, with a focus on evaluating energy efficiency and dynamic response. The system's energy equation can be expressed as

$$E_{total} = \frac{1}{2} m_{eff} \dot{y}_G^2 + \frac{1}{2} k_1 (\Delta L_1)^2 + \frac{1}{2} k_2 (\Delta L_2)^2 \quad (8)$$

Where m_{eff} is the equivalent mass.

The dynamic characteristics are described by the stiffness matrix, and the system's equivalent stiffness k_{eff} is

$$k_{eff} = \frac{dF_{ext}}{dy_G} = k_1 \left(\frac{\partial(\Delta L_1)}{\partial y_G} \right)^2 + k_2 \left(\frac{\partial(\Delta L_2)}{\partial y_G} \right)^2 \quad (9)$$

The stiffness value varies with θ , increasing significantly during the contact phase.

The overall performance analysis demonstrates that the DS Hand exhibits outstanding energy efficiency, attributable to the minimal energy dissipation of its rigid structure. The equivalent stiffness formula reveals that the system possesses variable stiffness characteristics, ensuring stable support for the DS Hand.

IV. EXPERIMENTAL VALIDATION

To evaluate the performance of the proposed design, a prototype of the DS Hand was fabricated. With the exception of the metal motors and bearings, most components were produced by 3D printing using PLA material, while the finger plate patches were made from 2 mm thick silicone material. After assembly and calibration, a series of systematic experiments were conducted to validate its ability to grasp objects of various sizes and scoop cards.

A. Environmental Adaptability

To verify the strong environmental adaptability of the DS Hand, a downward pressing test was conducted. In the experiment, the DS Hand, serving as the end-effector, was pressed downward onto the tabletop. Initially, a small downward force was applied. Through the passive compliance of the dual linear rails and springs, the DS Hand absorbed vertical displacement of more than 20 mm during the pressing process (Fig. 7). A larger downward force was then applied, causing the hand to reach its displacement threshold. The vertical displacement of the hand reached its limit, and during the second downward pressing, the hand absorbed more than 55 mm of vertical displacement (Fig. 7). This fully demonstrates its strong environmental adaptability, while also preventing structural damage caused by excessive downward force during robotic arm operations, thereby enhancing stability.

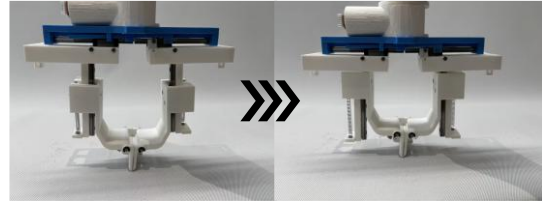


Fig. 7. Environmental adaptability test

B. Adaptive Grasping

Based on the above adaptability tests, the grasping ability of the DS Hand was further evaluated. Three typical daily objects were selected: a small, smooth, elongated object (metal shaft), a slightly larger irregular object (pen cap), and a wide, smooth-surfaced cylindrical object (plastic bottle). The results showed that the DS Hand was able to grasp these objects very stably, completely lifting them off the surface with the robotic arm, facilitating subsequent operations, as shown in Fig. 8.

C. Adaptive Switching Between Grasping and Scooping Modes

The DS Hand can automatically switch between grasping and scooping. When the manipulator presses downward to a preset depth, it causes the DS Hand to tilt, actively adjusting its grasping posture. At this stage, the two fingers experience uneven forces, and the mechanism enters an asymmetric card-scooping mode. The fingertip on the side with the smaller angle

performs the scooping action, while the side with the larger angle acts as a pivot, causing the card to flip and increasing the contact area with the fingertips. Experimental results show that the DS Hand achieves a high success rate when scooping thin and lightweight objects such as playing cards (Fig. 9), fully demonstrating its specialized adaptability to thin objects. After the scooping process is completed, the manipulator lifts upward, and under the combined effect of the dual linear rails and springs, the DS Hand returns to the symmetric grasping mode, reflecting its adaptive switching capability between the grasping and card-scooping modes.



Fig. 8. Adaptive grasping test

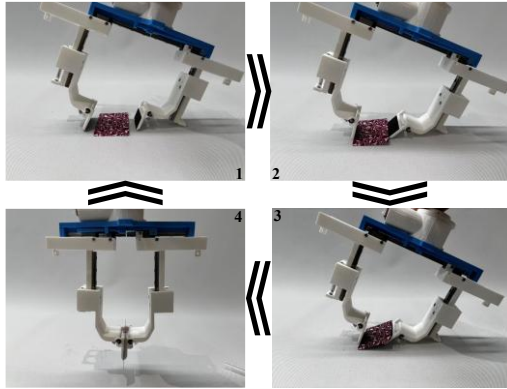


Fig. 9. The process of card scooping

V. CONCLUSIONS

This paper presents the design and analysis of the DS Hand, a novel underactuated robotic hand mechanism that combines a dual-spring mechanism with a sliding guide system. The hand utilizes a combination of springs and a fixed linkage structure to achieve high adaptability and efficiency in grasping and scooping tasks. By integrating the springs with a specific linkage system, the DS Hand demonstrates strong environmental interaction capability and grasping stability, while being able to handle objects of various shapes and sizes. A major advantage of the design lies in its simplicity and strong environmental adaptability: the main structure of the hand consists of two sliding rails and springs, making it simpler than other mechanisms while offering the same grasping and scooping functionalities, with the added benefit of adaptive switching. Additionally, it has stronger environmental adaptability,

enhancing the DS Hand's performance across a wide range of applications. Future research will focus on adding control modules to the hand, optimizing its material properties, and expanding its capabilities for more complex grasping and manipulation tasks. Overall, the DS Hand provides an innovative solution for robotic manipulation, combining simplicity and efficiency, with great potential for practical application in industrial fields such as automation and assistive robotics.

REFERENCES

- [1] A. Bicchi "Hands for Dexterous Manipulation and Robust Grasping: A Difficult Road Toward Simplicity," *IEEE Trans. on Robotics*, vol. 16, no. 6, pp. 652–662, 2000.
- [2] S. Jacobsen, E. Iversen, D. Knutti et al., "Design of the Utah/M.I.T. Dexterous Hand," *IEEE Inter. Conf. on Robotics and Automation Proceedings*, Apr. 1986, pp. 1520–1532.
- [3] H. Liu et al., "Multisensory Five-Finger Dexterous Hand: The DLR/HIT Hand II," *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, Nice, France, 2008.
- [4] C. Loucks, V. Johnson, P. Boissiere, et al., "Modeling and Control of the Stanford/JPL hand," *Proc. IEEE Int. Conf. on Robotics and Automation*, Raleigh, NC, USA, 1987.
- [5] L. Birglen, C. M. Gosselin, "Kinestatic Analysis of Underactuated Fingers," *IEEE Trans. on Robotics*, vol. 20, no. 2, pp. 211–221, Apr. 2004.
- [6] Townsend W. "The Barrett Hand Grasper-Programmably Flexible Part Handling and Assembly." *Industrial Robot: An Int. J.* 2000, 27(3): 181–188.
- [7] Martin E., Desbiens A. L., Lalibert E. T., et al., "SARAH Hand Used for Space Operations on STVF Robot," 2004 Int. Conf. on Intelligent Manipulation and Grasping, Genoa, Italy, July 1-2, 2004: 325-331.
- [8] E. Brown, N. Rodenberg, J. Amend, et al., "Universal Robotic Gripper Based on the Jamming of Granular Material," *Proc. of the National Academy of Sciences*, vol. 107, no. 44, pp. 18809–18814, 2010.
- [9] Y. Okatani, T. Nishida, K. Tadakuma, "Development of Universal Robot Gripper Using MR α Fluid," *Proc. of SCIS & ISIS 2014*, Kitakyushu, Japan, Dec. 3–6, 2014.
- [10] J. Xu et al., "LPSA underactuated mode of linearly parallel and selfadaptive grasping in the CLIS robot hand with Chebyshev linkage and idle stroke," 2017 2nd Inter. Conf. on Advanced Robotics and Mechatronics (ICARM), Hefei and Tai'an, China, 2017.
- [11] Y. Liu, W. Zhang, "An Underactuated Robot Finger with Movable Skeleton Link and Hoecken Mechanism for Linear Clamping and Self-adaptive Grasping," *IEEE Int. Conf. on Real-time Computing and Robotics (RCAR)*, Irkutsk, Russia, 2019.
- [12] J. Calderón, I. Juárez, J. Anzures et al., "Analysis and synthesis of the Peaucellier mechanism," *IEEE Inter. Autumn Meeting on Power, Electronics and Computing (ROPEC)*, Nov. 2016, pp. 1–6.
- [13] D. Yoon, K. Kim, "Fully Passive Robotic Finger for Human-Inspired Adaptive Grasping in Environmental Constraints," *IEEE/ASME Trans. on Mechatronics*, vol. 27, pp. 3841–3852, 2022.
- [14] D. Yoon, Y. Choi, "Analysis of fingertip force vector for pinch-lifting gripper with robust adaptation to environments," *IEEE Trans. Robot.*, vol. 37, no. 4, pp. 1127–1143, Aug. 2021.

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